PROJECT MANAGEMENT

One of the main reasons for exposure to project management is to provide young undergraduates with district advantages when it comes time to look for jobs.
More and more employers are looking fir graduates with project management skills. The logical starting point for developing these skills is understanding the UNIQUENESS of a project and of project managers.
What do the following headlines have in common?

(i) Citywide Wi-Fi system set to go live.
(ii) Apple's new iPhone hits the market.
(iii) Hosting of the 2022 world cup
(iv) Construction and opening of the 4th mainland bridge, or ShopRite mall.
(v) 1000-acre wind farm turns on the juice.

All of these events represent PROJECTS definition.
A project is a temporary endeavor undertaken to create a unique product service or result.
The major goal of a project is to satisfy a customer's needs.

The major characteristics of a project are as follows:

1. An established objective
2. A defined life span with a beginning and an end.
3. Usually the involvement of several departments and professionals.
4. Typically, doing something that has never been done before.
5. Specific time, cost and performance objectives/requirements.

WHAT A PROJECT IS NOT

- Projects should not be confused with everyday work.
- A project is not routine, repetitive work.
- Ordinary daily work typically requires doing the same or similar workover and over, while a project is done only once. e.g. a new product or service exists when the project is completed.

The table below examines the list that compares routine, repetitive work with projects;

s/n	Routine repetitive work	Projects
1.	Taking class note.	Writing a term paper.
2.	Daily entering sales receipts into the accounting ledger.	Setting up sales kiosk for a professional accounting meeting.
3.	Practicing scales on a piano	Writing a new piano piece
4.	Attaching tags on a manufactured product	Wire-tag projects for General Electric and Walmart.
5.	Routine manufacture of an Apple iPod	Designing an iPod that is approximately 2x4 inches, interfaces with pc and stores 10,000 songs.

AN OVERVIEW OF PROJECT MANAGEMENT

WHAT IS PROJECT MANAGEMENT?

A team-based approach for managing projects.

HOW IS IT DIFFERENT FROM GENERAL OPERATIONS MANAGEMENT

(i) Limited time frame
(ii) Narrow focus, specific objectives
(iii) Less bureaucratic

WHY IS IT USED

(i) Special needs that don't lend themselves to functional management.
(ii) Pressures for new or improved products or services, and cost reduction.

WHAT ARE THE KEY METRICS?

(i) Time
(ii) Cost
(iii) Performance objectives

WHAT ARE THE KEY SUCCESS FACTORS?

(i) Top-down commitment
(ii) A respected and capable project manager.
(iii) Enough time to plan
(iv) Careful tracking and control
(v) Good communications

WHAT ARE THE MAJOR ADMINISTRATIVE ISSUES?

1. Executive responsibilities
    a. Project selection
    b. Selecting of a project manager.
    c. Organizational structure (to whom will the project manager report).
2. Organizational alternatives:
    a. Manage with functional unit
    b. Assign a coordinator
    c. Use a matrix organization with a project leader.

WHAT ARE THE MAIN TOOLS?

1. **Work breakdown structure:** An initial planning tool that is needed to develop a list of activity sequences and a realistic budget.
2. **Network diagram:** A "big picture" visual aid that is used to estimate project duration, identify activities that are critical for timely project completion, identify areas where slack time exists, and develop activity schedules.
3. **Gantt charts**: A visual aid used to plan and monitor individual activities.
4. **Risk management**: Analysis of potential failures or problems, assessment of their likelihood and consequences and contingency plans.

KEY DECISIONS IN PROJECT MANAGEMENT

Much of the success of projects depend on

(i) Deciding which projects to implement: This involves determining the criteria that will be used, factors include budget, availability of appropriate knowledge and skill personnel, cost-benefit considerations.

(ii) Selecting the project manager: This is the control person in the project. He is responsible for managing:
    a. The work
    b. Human resources – those working have direction and motivation
    c. Communications: everybody has the information needed to do the work.
    d. Quality
    e. Time: Project is completed on schedule.
    f. Costs: completed within budget.

    The position of project manager has high visibility.

(iii) Selecting the project team: how will the person work with others.

(iv) Planning and designing the project: requires decisions on project performance goals,

   a. Timetable for prompt completion, the scope of the project, what work needs to be done, how it will be done, if some portions will be outsourced, a budget and what resources will be needed.

(v) Managing and controlling project resources: involves managing personnel, equipment and the budget.

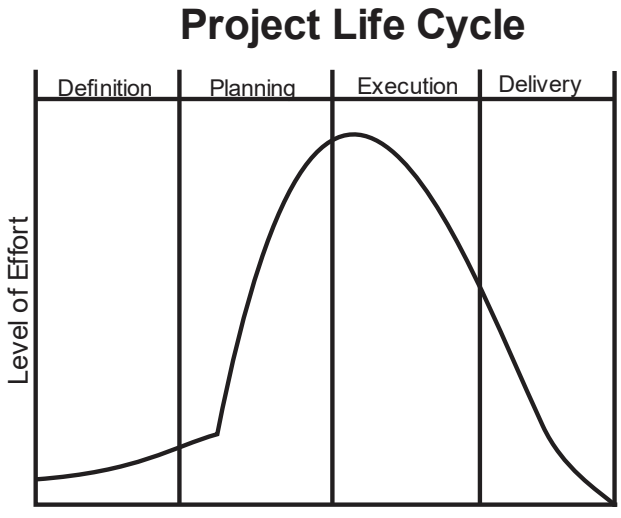
(vi) Deciding if and when a project should be terminated: important, consideration is likelihood of success, termination costs and whether resources could be better used elsewhere.

PROJECT LIFE CYCLE

Projects go through a life cycle which typically consists of four phases:

1. Definition:
   a. Concept: the organization recognizes the need for a project, responds to a request for a proposal from a client.
   b. Feasibility analysis: examines the expected costs, benefits and risks of undertaking the project.
2. Planning: This spells out the details of the work and provides estimates of the necessary human resources, time and costs.
3. Execution: Here, the project is done. This phase accounts for the majority of time and resources consumed by a project.
4. Termination: Closure is achieved. Deals with leftover, materials, equipment and any other resources associated with the project.

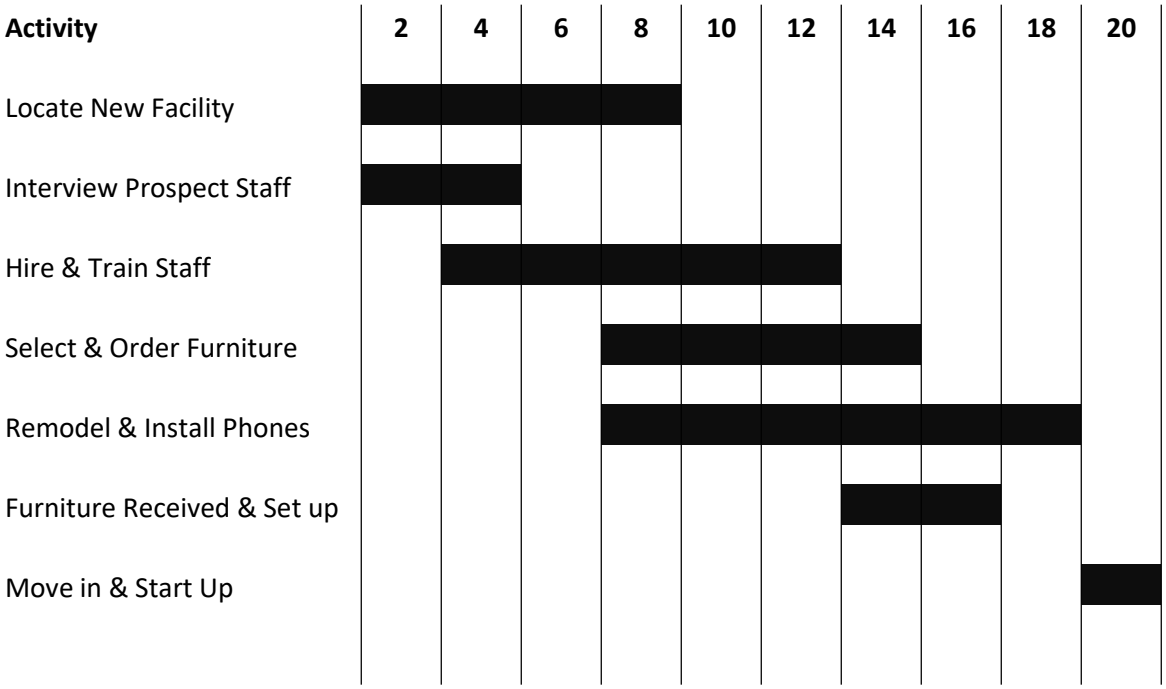
It should be noted that phases can overlap. This can reduce the time necessary to move through the life cycle and therefore generating cost savings.



1. Goals	1. Schedules	1. Status Reports	1. Train Consumer
2. Specifications	2. Budgets	2. Changes	2. Transfer Documents
3. Feasibility	3. Resources	3. Quality	3. Release Resources
4. Tasks	4. Risks		4. Reassign staff
5. Responsibilities	5. Staffing		5. Lessons learned
6. Teams			

PLANNING AND SCHEDULING WITH GANTT CHARTS

The Gantt chart is a popular tool for planning and scheduling simple projects. It enables a manager to initially schedule project activities and then to monitor progress over time by comparing planned progress to actual progress.
Its advantage is that it is simple.



PERT AND CPM

PERT (PROGRAM EVALUATION AND REVIEW TECHNIQUE) and CPM (CRITICAL PATH METHOD) are the two most widely used techniques for planning and coordinating large-scale projects.

By using these 2 techniques, managers are able to obtain:

(i) A graphical display of project activities.
(ii) An estimate of how long the project will take.
(iii) An indication of which activities are the most critical to timely project completion.
(iv) An indication of how long any activity can be delayed without delaying the project.

NETWORK DIAGRAM
Diagram of project activities that shows sequential relationships by use of arrows and nodes.

Activity-on-Arrow (AOA)
Network diagram convention in which arrows designate activities.

ACTIVITY-ON-NODE (AON)
Network diagram convention in which nodes designate activities.

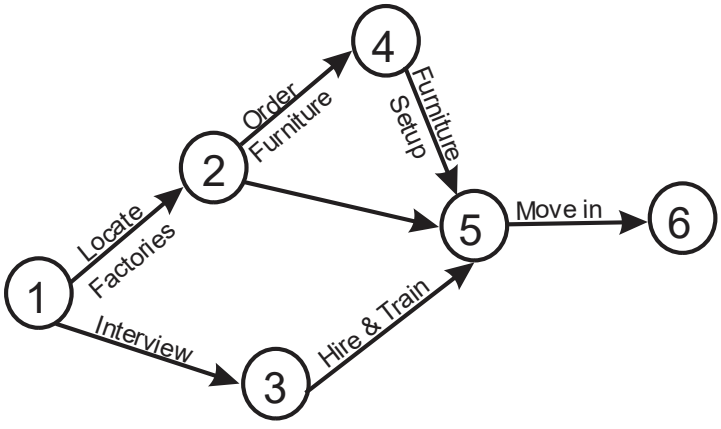
PATH
A sequence of activities that leads from the starting node to the finishing node.

CRITICAL PATH
The longest path; determines expected project duration.

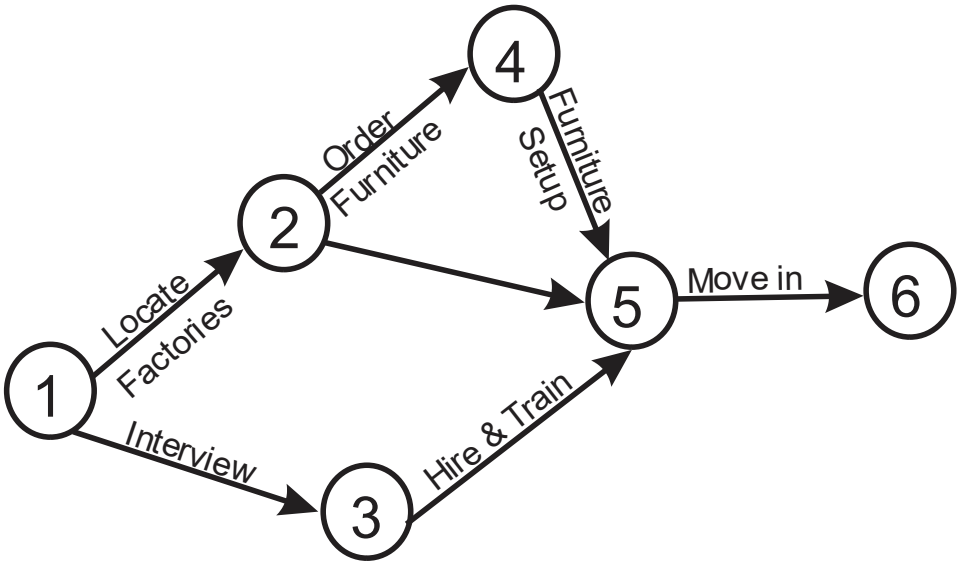
CRITICAL ACTIVITIES
Activities on the critical path.

SLACK
Allowable slippage for a path, the difference between the length of a path and the length of the critical path.

Activity-on-Arrow (AOA) Diagram



Activity-on-Node (AON) Diagram

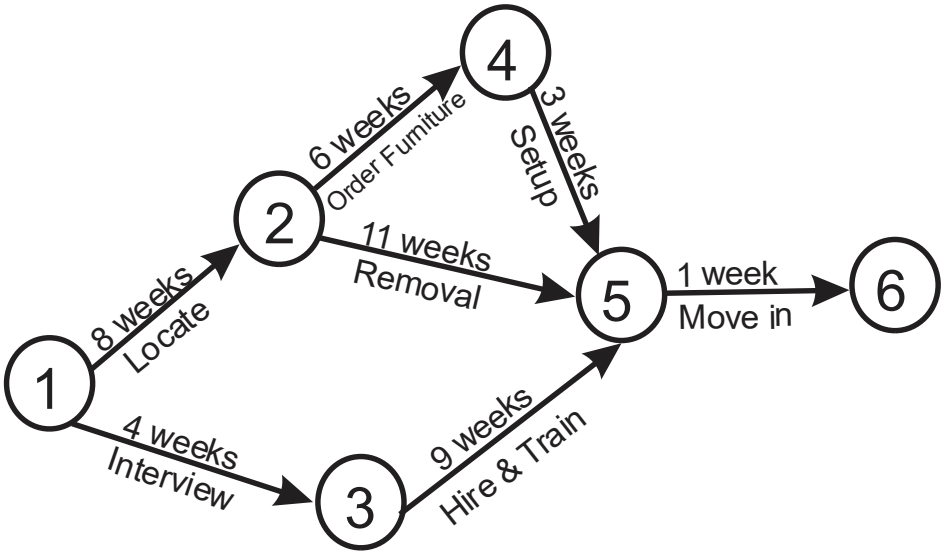


Example

Given the additional information on the bank network below, determine

(a) The length of each path
(b) The critical path
(c) The expected length of the project
(d) The amount of slack time for each path.

(AOA Diagram)



(a)

(i) Path 1 – 2 – 4 – 5 – 6 = 8 + 6 + 3 + 1 = 18weeks
(ii) Path 1 – 2 – 5 – 6 = 8 + 11 + 1 = 20 weeks
(iii) Path 1 – 3 – 5 – 6 = 4 + 9 + 1 = 14 weeks

(b) Path 1 – 2 – 5 – 6 is the longest path (20 weeks), so it is the critical path.

(c) The expected length of the project is equal to the length of the critical path (i.e., 20 weeks)

(d) We find the slack for each path by subtracting its length from the length of the critical path

Path	**Length (Wks)**	**Slack (Wks)**
1 -2 – 4 – 5 – 6	8 + 6 + 3 + 1 = 18	20 – 18 = 2
1 – 2 – 5 – 6	8 + 11 + 1 = 20	20 – 20 = 0
1 – 3 – 5 – 6	4 + 9 + 1 = 14	20 – 14 = 6